

UNIVERSITY OF ADEN
FACULTY OF ENGINEERING
TEST OF EVEN SEMESTER FOR ACADEMIC YEAR (2025/2026)

Subject :Data Structure

DATE : 06 /05 /2026

Class :B2IT/E

Student Name :

Reg.No:.....

Q1:What will be the output of the programs?	Q2:Convert the infix expression to postfix
<pre>#include <iostream> #include <queue> using namespace std; int main() { queue < int >Q; for (int i = 0; i <=5; i++) { Q.push(i*3); } cout << "The size of queue is " << Q.size() << '\n'; cout << "The firts element in the queue is : " << Q.front() << '\n'; cout << "The elements of the queue are: "; while (!Q.empty()) { cout << Q.front()<< ' '; Q.pop(); } return 0; }</pre>	A*B*C+D
	A+B ^D
The output is :	

Q3:Answer the question

- 1- What is a queue?.....
- 2- What does the "top" refer to in a satch?.....
- 3- A stack follows
- 4- List two real-life examples of a stack implementation?.....
- 5- How the pop() operation modifies the top() in stack?.....
- 6- The process of inserting element to a queue is also called.....
- 7- What is a static data structure?.....
- 8- What is the result of a performing a front() operation on a queue that contains 30 , 20, 40 elements.....
- 9- What is the mistake:
Stack<int>,
Cout<<st.top();
- 10-write the syntax to declare a queue in c++?.....

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Q1:What will be the output of the programs?

```
#include <iostream>
#include <queue>
using namespace std;
int main() {
    queue < int >Q;
    for (int i = 0; i <=9; i++) {
        Q.push(i + 3);
    }
    cout << "The size of queue is " << Q.size() << "\n";
    cout << "The firts element in the queue is : " << Q.front() <<
    '\n';
    cout << "The elements of the queue are: ";
    while (!Q.empty()) {
        cout << Q.front() << ' ';
        Q.pop();
    }
    return 0;
}
```

The output is :

Q2:Convert the infix expression to postfix

A+B / (C+D)

A* B^C^D

Q3: Answer the question

- 1- What does the "Front" refer to in a queue?.....
- 2- With every pop() in the stack the top is.....
- 3- How do you check if a stack is full?.....
- 4- How the push() operation modifies the top() in stack?.....
- 5- The process of inserting element in a queue is also called.....
- 6- If the top variable is greater then or equal to size -1 then the stack is
- 7- What is the result of a performing a pop() operation on a stack that contains 100,200,400 elements.....
- 8- Give two examples of non-linear data structures?.....
- 9- What type of queue allows elements to be added or removed from both ends?.....
- 10- Where does a stack add new items?.....

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Q1: What will be the output of the programs?

```
#include <iostream>
#include <stack>
using namespace std;

int main() {
    stack < int > s;
    for (int i = 0; i < 5; i++) {
        s.push(i + 2);
    }
    cout << "The size of stack is " << s.size() << '\n';
    cout << "The topmost element of the stack is " << s.top() << '\n';
    s.pop();
    s.pop();
    cout << "The elements of the stack after pop operation ";
    while (!s.empty()) {
        cout << s.top() << ' ';
        s.pop();
    }
    return 0;
}
```

The output is :

Q2: Convert the infix expression to postfix

$(A+B) * (C-D)$

$A + B / (C - D^E)$

Q3: Answer the question

- 1- How to get the top element in stack ?
- 2- What is the function used to insert an element in a queue?
- 3- What are the types of data structure?
- 4- In which condition the queue is full ?
- 5- What does empty() check in queue?
- 6- Give one example of non-linear data structure?
- 7- Which functions are used to access the elements of queue?
- 8- Which data structure uses FIFO?
- 9- What is the mistake?

top = top - 1

stack[top] = value;

- 10- If queue contains 5, 10, 15, what is the rear(back) element ?

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Q1:What will be the output of the programs?

Q2:Convert the infix expression to prefix

```
#include <iostream>
#include <stack>
using namespace std;
int main() {
    stack < int > s;
    for (int i = 0; i <=5; i++) {
        s.push(i * 2);
    }
    cout << "The size of stack is " << s.size() << '\n';
    cout << "The top element of the stack is " << s.top() << '\n';
    cout << "The elements of the stack are: ";
    while (!s.empty()) {
        cout << s.top() << ' ';
        s.pop();
    }
    return 0;
}
```

$(A * B - C) / D$

$A - B^{\wedge}C^{\wedge}D$

The output is :

Q3:Answer the question

- 1- What is a stack?.....
- 2- What does the "Rear" refer to in a queue?.....
- 3- A queue follows
- 4- List two real-life examples of a queue implementation?.....
- 5- How the pop() operation modifies the top() in stack?.....
- 6- The process of deleting element from a queue is also called.....
- 7- What is a dynamic data structure?.....
- 8- What is the result of a performing a pop() operation on a queue that contains 30 , 20, 40 elements.....
- 9- What is the mistake:
 $queue <int> q;$
 $q.pop();$
- 10-write the syntax to declare a stack in c++?.....

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Q1:What will be the output of the programs?

```
#include <iostream>
#include <queue>
using namespace std;

int main() {
    queue < int > s;
    for (int i = 0; i <= 9; i++) {
        s.push(i + 1);
    }
    cout << "The size of queue is " << s.size() << '\n';
    cout << "The front element of the queue is " << s.front() << '\n';
    cout << "The rear element of the queue is " << s.back() << '\n';
    s.pop();

    cout << "The elements of the queue after pop operation ";
    while (!s.empty()) {
        cout << s.front() << ' ';
        s.pop();
    }
    return 0;
}
```

The output is :

Q2:Convert the infix expression to postfix

$(A+B) * (C * D - E)$

$(C - D^E) / (A + B)$

Q3:Answer the question

- 1- With every push in the stack, the top is
- 2- The return value of **top()** function after adding "S","A","M","D" and delete two elements from stack is
- 3- **LIFO** data structure forms for :
- 4- How do you check if a stack is empty?.....
- 5- How the pop() operation modifies the top() in stack?.....
- 6- The process of inserting element in a queue is also called.....
- 7- If the top variable is greater then or equal to size -1 then the stack is
- 8- What is the result of a performing a pop operation on a stack that contains elements 10,20,30.....
- 9- Give two examples of linear data structures?.....
- 10- What is a static data structure?.....

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Q1:What will be the output of the programs?	Q2:Convert the infix expression to postfix
<pre>#include <iostream> #include <stack> using namespace std; int main() { stack < int > s; for (int i = 0; i < 5; i++) { s.push(i + 1); } cout << "The size of stack is " << s.size() << '\n'; cout << "The topmost element of the stack is " << s.top() << '\n'; s.pop(); s.pop(); cout << "The elements of the stack after pop operation "; while (!s.empty()) { cout << s.top() << ' '; s.pop(); } return 0; }</pre>	A+B*C-D
	A - B / (C*D^E)
The output is :	

Q3:Answer the question

- 1- What does **FIFO** mean?.....
- 2- Which operations modifies the stack?.....
- 3- What are the two types of non-primitive data structure?.....
- 4- In which condition the queue is full ?.....
- 5- Popping an element from an empty stack is called
- 6- Give one example of non-linear data structure?.....
- 7- Which function is used to access the top element of stack?.....
- 8- Complete : Algorithm + Data Structure =
- 9- What is the mistake?.....
- (if top==size)
- cout<<"full";
- 10- In a queue ,where are elements inserted and deleted?.....